Library Management Interface — JavaScript Code

Save this code in a file named **script.js**. In index.html, include `<script src="script.js"></script>` just before the closing body tag.

```
const defaultBooks = [
  { id: 1, title: "The Alchemist", author: "Paulo Coelho", category: "Fiction", year: 1988, status: "Available", borrower: "" },
  { id: 2, title: "Clean Code", author: "Robert C. Martin", category: "Technology", year: 2008, status: "Issued", borrower: "Arun" }
];

function loadBooks() {
  try {
    const savedBooks = localStorage.getItem("libraryBooks");
    return savedBooks === null
      ? defaultBooks.map(book => ({ ...book }))
      : JSON.parse(savedBooks);
  } catch (error) {
    return defaultBooks.map(book => ({ ...book }));
  }
}

let books = loadBooks();

function saveBooks() {
  localStorage.setItem("libraryBooks", JSON.stringify(books));
}

function escapeHTML(value) {
  return String(value ?? "").replace(/[&<>"']/g, character => ({
    "&": "&amp;",
    "<": "&lt;",
    ">": "&gt;",
    '"': "&quot;",
    "'": "&#39;"
  })[character]);
}

function updateStats() {
  document.getElementById("totalBooks").textContent = books.length;
  document.getElementById("availableBooks").textContent =
    books.filter(book => book.status === "Available").length;
  document.getElementById("issuedBooks").textContent =
    books.filter(book => book.status === "Issued").length;
}

function renderBooks() {
  const searchText = document.getElementById("search").value.toLowerCase().trim();
  const selectedStatus = document.getElementById("statusFilter").value;

  const filteredBooks = books.filter(book => {
    const title = String(book.title || "").toLowerCase();
    const author = String(book.author || "").toLowerCase();
    const category = String(book.category || "").toLowerCase();

    const matchesSearch =
      title.includes(searchText) ||
      author.includes(searchText) ||
      category.includes(searchText);

    const matchesStatus =
      selectedStatus === "" || book.status === selectedStatus;

    return matchesSearch && matchesStatus;
  });

  const tableBody = document.getElementById("bookTable");

  if (filteredBooks.length === 0) {
    tableBody.innerHTML = '<tr><td colspan="4">No books found.</td></tr>';
  } else {
    tableBody.innerHTML = filteredBooks.map(book => `
      <tr>
        <td>
          <strong>${escapeHTML(book.title)}</strong><br>
          ${escapeHTML(book.author)}
          ${book.year ? "<br>" + escapeHTML(book.year) : ""}
          ${book.borrower ? "<br>Borrower: " + escapeHTML(book.borrower) : ""}
        </td>
        <td>${escapeHTML(book.category)}</td>
        <td>${escapeHTML(book.status)}</td>
        <td>
          <button class="secondary" type="button" onclick="editBook(${Number(book.id)})">Edit</button>
          <button class="primary" type="button" onclick="toggleBook(${Number(book.id)})">
            ${book.status === "Available" ? "Issue" : "Return"}
          </button>
          <button class="danger" type="button" onclick="deleteBook(${Number(book.id)})">Delete</button>
        </td>
      </tr>
    `).join("");
```

```
  }

  updateStats();
}

document.getElementById("bookForm").addEventListener("submit", function(event) {
  event.preventDefault();

  const bookId = document.getElementById("bookId").value;
  const bookData = {
    title: document.getElementById("title").value.trim(),
    author: document.getElementById("author").value.trim(),
    category: document.getElementById("category").value,
    year: document.getElementById("year").value
  };

  if (!bookData.title || !bookData.author || !bookData.category) {
    alert("Please fill in the required book details.");
    return;
  }

  if (bookId) {
    const existingBook = books.find(book => book.id === Number(bookId));
    if (existingBook) {
      Object.assign(existingBook, bookData);
    }
  } else {
    books.push({
      id: Date.now(),
      ...bookData,
      status: "Available",
      borrower: ""
    });
  }

  saveBooks();
  cancelEdit();
  renderBooks();
});

function editBook(id) {
  const book = books.find(item => item.id === id);
  if (!book) return;

  document.getElementById("bookId").value = book.id;
  document.getElementById("title").value = book.title;
  document.getElementById("author").value = book.author;
  document.getElementById("category").value = book.category;
  document.getElementById("year").value = book.year || "";
  document.getElementById("formTitle").textContent = "Edit Book";
  document.getElementById("submitBtn").textContent = "Save Changes";

  window.scrollTo({ top: 0, behavior: "smooth" });
}

function cancelEdit() {
  document.getElementById("bookForm").reset();
  document.getElementById("bookId").value = "";
  document.getElementById("formTitle").textContent = "Add a Book";
  document.getElementById("submitBtn").textContent = "Add Book";
}

function deleteBook(id) {
  const book = books.find(item => item.id === id);
  if (!book) return;

  if (confirm("Delete " + book.title + "?")) {
    books = books.filter(item => item.id !== id);
    saveBooks();
    renderBooks();
  }
}

function toggleBook(id) {
  const book = books.find(item => item.id === id);
  if (!book) return;

  if (book.status === "Available") {
    const borrowerName = prompt("Enter borrower's name:");
    if (borrowerName === null) return;

    if (!borrowerName.trim()) {
      alert("Please enter the borrower's name.");
      return;
    }

    book.status = "Issued";
    book.borrower = borrowerName.trim();
  } else {
    book.status = "Available";
    book.borrower = "";
  }
```

```
  saveBooks();
  renderBooks();
}

function resetBooks() {
  if (confirm("Reset the book list to demo data? This will replace your current list.")) {
    books = defaultBooks.map(book => ({ ...book }));
    saveBooks();
    cancelEdit();
    renderBooks();
  }
}

renderBooks();
```